center of a black hole (Chapter 13). You are preaching to physicists that the endpoint of stellar implosion is a great crisis, from which deep new understanding may emerge. You are not preaching to astronomers that they should search for black holes, or even neutron stars. Of searches for black holes you say nothing; of the more promising idea to search for a neutron star, you echo in your writings the conservative view of the astronomical community: "Such an object will have a diameter of the order of 30 kilometers. . . . it will cool rapidly. . . . There is about as little hope of seeing such a faint object as there is of seeing a planet belonging to another star" (in other words, no hope at all).

Imagine yourself as Yakov Borisovich Zel'dovich. It is 1964; Mikhail Podurets, a member of your old hydrogen bomb design team, has just finished his computer simulations of stellar implosion including the effects of pressure, shock waves, heat, radiation, and mass ejection (Chapter 6). The simulations produced a black hole (or, rather, a computer's version of one). You are now fully convinced that some massive stars, when they die, must form black holes. Do you next sit down with astronomers and plan a search for them? Yes, by all means. If you are Zel'dovich, then you have little sympathy for Wheeler's obsession with the endpoint of stellar implosion. The endpoint will be hidden by the hole's horizon; it will be invisible. By contrast, the horizon itself and the hole's influence on its surroundings might well be observable; you just need to be clever enough to figure out how. Understanding the observable part of the Universe is your obsession, if you are Zel'dovich; how could you possibly resist the challenge of searching for black holes?

Where should your search begin? Clearly, you should begin in our own Milky Way galaxy—our disk-shaped assemblage of 10^{12} stars. The other big galaxy nearest to our own, Andromeda, is 2 million light-years away, 20 times farther than the size of the Milky Way; see Figure 8.1. Thus, any star or gas cloud or other object in Andromeda will appear 20 times smaller and 400 times dimmer than a similar one in the Milky Way. Therefore, if black holes are hard to detect in the Milky Way, they will be 400 times harder to detect in Andromeda—and enormously harder still in the 1 billion or so large galaxies beyond Andromeda.